

# RUPAYAN CHATTARAJ

Computer Science & Engineering Student

+91 8918158588 | rupayanchattaraj@gmail.com | Github | Portfolio | Vellore, Tamil Nadu, India

## Professional Summary

Computer Science student and self-taught full-stack developer with hands-on experience building AI-integrated web applications, computer-vision systems, and backend services in Python, Flask, and JavaScript. Skilled at integrating third-party APIs (Groq, Gemini) into production-style applications and deploying full projects end-to-end. Strong foundation in Data Structures and Algorithms, actively developed through consistent LeetCode practice.

## Interests

Solo Traveling | Competitive Sports | Competitive Programming | Independent Software Projects

## Education

**Vellore Institute of Technology (VIT), Vellore**

*Expected Graduation: 2028*

B.Tech in Computer Science and Engineering – CGPA: 8.74/10 (through 4th semester)

**DAV Model School, Durgapur, West Bengal**

*Completed: 2024*

Higher Secondary Education (Class XII) – Grade: 89%

## Technical Skills

**Languages:** Java, Python, C, C++, JavaScript

**Frontend Development:** HTML, CSS, JavaScript, React.js

**Backend Development:** Node.js, Flask, Email.js, REST APIs

**Databases:** MySQL, Oracle Database 21c (SQL\*Plus, PL/SQL basics)

**AI & Computer Vision:** OpenCV, MediaPipe, Groq API, Gemini API

**Tools & Platforms:** Git, GitHub, VS Code, Postman, Render, Netlify

**Core Concepts:** Data Structures & Algorithms, RESTful API Design, Full-Stack Architecture

## Projects

**Rusho.Bot – AI-Powered Virtual Assistant**

*Python, Flask, Groq API, Gemini API, JavaScript*

- Engineered a full-stack AI assistant with a Flask backend integrating Groq (Llama 3.3 70B, Llama 3.1 8B) and Gemini APIs, supporting text and voice interaction.
- Designed ten dedicated AI tool modules – including a Resume Analyzer, Code Reviewer, Quiz Generator, PDF Chat, and Translator – each with independent routes and tailored system prompts.
- Built a glassmorphic chat widget with model selection, auto-expanding input, and real-time response handling; deployed publicly on Render.

**Wavify – Music Streaming Web Application**

*HTML, CSS, JavaScript*

- Developed a responsive music streaming interface with playlist management and full playback controls.
- Built a modern, component-driven UI focused on usability and visual polish.

**Hand Gesture Controller – Touchless Input System**

*Python, OpenCV, MediaPipe*

- Built a real-time hand-tracking system enabling touchless computer interaction using computer vision.
- Improved gesture recognition accuracy through refined MediaPipe landmark processing.

**Eye Pointer – Eye-Tracking Cursor Control**

*Python, OpenCV, MediaPipe*

- Developed an eye-tracking application enabling cursor control via facial landmark detection.
- Explored accessibility-focused human-computer interaction as a core design goal.

## Achievements & Continuous Learning

- Built and deployed multiple full-stack and AI-powered applications independently, from backend architecture through public deployment.
- Actively solving Data Structures & Algorithms problems on LeetCode to strengthen problem-solving fundamentals.
- Active participation in hackathons.
- Completed Certifications:** The Cybersecurity Threat Landscape; Microsoft Azure Fundamentals (AZ-900) Cert Prep: 1 Cloud Concepts (Microsoft Press); Ethical Hacking: Wireless Networks.